۲۸ دی ۱۴۰۴

جلسه TA یادگیری ماشین

1) Dimensionality Reduction $\longrightarrow$ PCA

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$$f : \mathbb{R}^d \to \mathbb{R}^K \quad : \quad \tilde{x} = f(x)$$

$$K \leq d$$

Cov Matrix

Pattern

Help: Dataset $\longrightarrow$ $X = [x^{(1)}, \dots, x^{(N)}]^T$

$$x^{(i)} \in \mathbb{R}^d$$

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PCA $\longrightarrow$ f is linear

$$f(x) = Ax, \quad \left.\begin{array}{l} A \in \mathbb{R}^{k \times d} \\ x \in \mathbb{R}^{d} \end{array}\right\} \; f(x) = Ax \in \mathbb{R}^{k}$$

Task : find the optimal A $\longrightarrow$ $Var(Ax)$ maximize

$\longrightarrow$ Reconstruction error minimize

$$x \xrightarrow{A} Ax \xrightarrow{A^T} \underline{\underline{A^T A x \approx x}}$$

$$\longrightarrow \|A^T A x - x\|_2^2$$

A is Projection matrix

$$AA^T = I$$

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$$X = [x^{(1)}, \dots, x^{(N)}]^T$$

$A \longrightarrow \mathrm{Var}(Ax)$ maximize $\qquad A \in \mathbb{R}^{K \times d}$

special case : $K = 1$ $\longrightarrow$ $A \in \mathbb{R}^{1 \times d} \rightsquigarrow \underline{\underline{v}}^T$

max $\mathrm{Var}(v^T x) = \mathbb{E}[(v^T x)^2] - \mathbb{E}[v^T x]^2$

$$= \frac{1}{N} \sum (v^T x^{(i)})^2 - \left(\frac{1}{n} \sum v^T x^{(i)}\right)^2$$

PCA : 1) zero-mean

$$x^{(i)} = [x_1^{(i)}, \dots, x_d^{(i)}]^T$$

} feature zero mean

$$\bar{x}_j = \frac{1}{N} \sum x_j^{(i)}$$

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$$\tilde{x}_j^{(i)} = x_j^{(i)} - \bar{x}_j$$

$$\longrightarrow \text{mean } \tilde{x}_j = 0$$

$$\mathrm{Var}(v^T x) = \underbrace{\frac{1}{N}\sum \left(v^T x^{(i)}\right)^2}_{} - \underbrace{\left(\frac{1}{N}\sum v^T x^{(i)}\right)^2}_{=0 \text{ (if zero mean)}}$$

$$\left(v^T x^{(i)}\right)^2 = v^T x^{(i)} x^{(i)T} v$$

$$\longrightarrow \mathrm{Var}(v^T x) = \frac{1}{N}\sum v^T x^{(i)} x^{(i)T} v = v^T \underbrace{\left(\frac{1}{N}\sum x^{(i)} x^{(i)T}\right)}_{} v$$

$$= \underbrace{v^T \Sigma v}_{}$$

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$$\max_{v} \; v^T \Sigma v \quad \text{s.t.} \quad \|v\|_2^2 = 1$$

$$\Sigma = \frac{1}{N} X X^T$$

$A \rightarrow$ Projection orthogonal

$$\mathcal{L}(v, \lambda) = v^T \Sigma v - \lambda(\|v\|_2^2 - 1) = v^T \Sigma v - \lambda v^T v + \lambda$$

$$\nabla_v \mathcal{L} = 0 \longrightarrow 2\Sigma v - 2\lambda v = 0 \longrightarrow \Sigma v = \lambda v$$

$$v^T \Sigma v = \lambda v^T v = \lambda$$

$\max \lambda$ s.t. λ is an eigenvalue of Σ

$$\longrightarrow \lambda = \lambda_{max}$$

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$v \longrightarrow$ eigenvector corresponding to biggest eigenvalue of Σ

Task : find $\overbrace{\text{optimal linear projection}}^{A}$ to K dims with dataset $X \in \mathbb{R}^{N \times d}$

$A = [v_1, \ldots, v_K]^T$, v_i is the eigenvector corresponding to the i'th largest eigenvalue of $\Sigma = \frac{1}{N} X X^T$

$v_i \in \mathbb{R}^d$

$A \in \mathbb{R}^{K \times d}$

$f(x) = Ax$

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PCA) given dataset $X = [x^{(1)}, \dots, x^{(N)}]^T \in \mathbb{R}^{N \times d}$

✓ 1) zero mean each feature

✓ 2) Compute Cov matrix $\frac{1}{N} X X^T \in \mathbb{R}^{d \times d}$

? 3) find eigenvalues of Σ and sort them

✓ 4) find top k eigenvectors $\rightarrow A = [v_1, \dots, v_k]^T$

5) $f(x) = Ax$

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PCA in Practice

1) Spectral decomp-- on $\Sigma = UDU^T$

$D = \mathrm{diag}(\lambda_1, \dots, \lambda_d)$

$U = [u_1, \dots, u_d]^T$

2) SVD on X

$X = U\Sigma' V^T$

$\frac{1}{N} XX^T = U\Sigma' V^T V \Sigma' U^T = U\Sigma'^2 U^T$

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8) Exam

$$\Sigma v = \lambda v \longrightarrow (\Sigma - \lambda I) v = 0$$

$$\longrightarrow \det(\Sigma - \lambda I) = 0 \longrightarrow \text{char... poly..}$$

$$\begin{bmatrix} a & b \\ b & c \end{bmatrix} \longrightarrow \begin{vmatrix} a-\lambda & b \\ b & c-\lambda \end{vmatrix} = 0$$

$$(\lambda - a)(\lambda - c) - b^2 = 0 \longrightarrow \lambda_1, \lambda_2$$

$$\Sigma v = \lambda_1 v \longrightarrow v^T v = 1$$

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$\longrightarrow \tilde{v}$ ✓

$$f(x) = \tilde{v}^T x$$

$$x \in \mathbb{R}^d \xrightarrow{W \in \mathbb{R}^{k \times d}} \underline{\underline{Wx}} \xrightarrow{U \in \mathbb{R}^{d \times k}} \underline{\underline{UWx}}$$

$k \leq d$

$$\min_{U,W} \sum_{i=1}^{N} \| x^{(i)} - UWx^{(i)} \|_2^2$$

$\longrightarrow$ U is orthogonal, $W = U^T$

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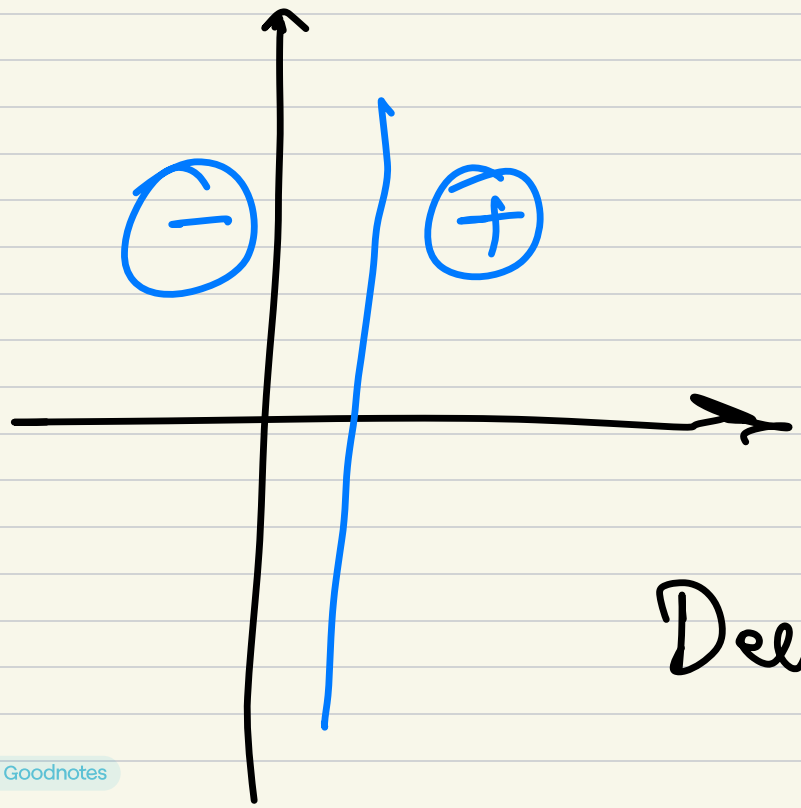
Ensemble Learning

Binary Classification → Strong Learner → KNN, SVM, logistic

Train (~ Test) error ↓

Binary Classification → Weak Learner

Test error $\leq 1/2 - \varepsilon$, $\varepsilon > 0$



Decision Stump

$\times N$ → Ensemble

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→ Bagging (Parallel)

→ Boosting (Sequential)

Bagging → Bootstrap + aggregating

↓ resample

↓ Voting, Average

$N \rightarrow x_1, \dots, x_N$

Bootstrap resampling : [illegible] N [illegible] → subsample

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Random Forest

T trees

d features

for each $t \in T$:

m feature $\underset{\text{random}}{\sim}$ d

$m \lessapprox \sqrt{d}$

decision tree $\rightsquigarrow$ m

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$\forall i \in [T]$, $h_i \longrightarrow$ ith decision tree

$$\mathrm{Var}(h_i) \leq \sigma^2$$

$$\mathrm{Correlation}(h_i, h_j) \leq \rho$$

$$\mathrm{Cov}(h_i, h_j) \leq \rho\sigma^2$$

$$H = \frac{1}{T}\sum h_i \longrightarrow \mathrm{Var}(H) = \frac{1}{T^2}\sum_i \mathrm{Var}(h_i)$$

$$+ \frac{1}{T^2}\sum_{i \neq j} \mathrm{Cov}(h_i, h_j)$$

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$$= \frac{\sigma^2}{T} + \frac{T(T-1)}{T^2}\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{T}\sigma^2$$

Adaboost

$h \in \mathcal{H} \longrightarrow$ function Class

M steps $\longrightarrow$ $h_1, \dots, h_M$

$\alpha_1, \dots, \alpha_M$

$$H(x) = \frac{1}{2}\left(\alpha_1 h_1(x) + \dots + \alpha_M h_M(x)\right)$$

$X = x^{(1)}, \dots, x^{(N)} \rightsquigarrow w^{(1)}, \dots, w^{(N)}$ (for one step)

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$$\text{loss}(h; X, w_j) = \sum_{i=1}^{N} w_j^{(i)} \times \mathbb{1}\left(y^{(i)} \neq h(x^{(i)})\right)$$

in step j

find h_j that minimizes $\text{loss}(h; X; w_j)$

1) how to set w_j ?

2) how to set α ?

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$$\varepsilon_j = \frac{\sum_{i=1}^{N} w_j^{(i)} \mathbb{1}\left(y^{(i)} \neq h_j(x^{(i)})\right)}{\sum w_j^{(i)}} \rightarrow \text{loss} = \sum \alpha$$

$$\sum w_j^{(i)} \rightarrow \text{normalizer} = \sum_{All}$$

$$\alpha_j = \ln\left(\frac{1-\varepsilon_j}{\varepsilon_j}\right)$$

$$w_1^{(i)} = \frac{1}{N}$$

$$w_{j+1}^{(i)} = w_j^{(i)} \exp\left\{\alpha_j \mathbb{1}\left(y^{(i)} \neq h_j(x^{(i)})\right)\right\}$$

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$\varepsilon_j \rightarrow$ loss of jth weak learner

$\varepsilon_j \downarrow \rightarrow \ln\left(\frac{1-\varepsilon_j}{\varepsilon_j}\right) \uparrow$

Algorithm)

1) init $w_1^{(i)} = 1/N$

2) for $j = 1, \dots, M$:

find h_j that minimizes loss

find ε_j, α_j, w_{j+1}

3) $H_M(x) = 1/2 \sum \alpha_j h_j(x)$

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Exp loss $\longrightarrow$ $\exp\{-y\hat{y}\}$

0-1 loss $\longrightarrow$ $\mathbb{1}(y \neq \hat{y})$

j-th step

$$H_j(x) = \frac{1}{2}\left[\alpha_1 h_1(x) + \cdots + \underline{\underline{\alpha_j h_j(x)}}\right]$$

?

$\alpha_1, \ldots, \alpha_{j-1}$ ✓

$h_1, \ldots, h_{j-1}$ ✓

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$$\text{Exp loss} = \sum_{i=1}^{N} e^{-y^{(i)} H_j(x^{(i)})}$$

$$= \sum e^{-y^{(i)}\left(\underbrace{H_{j-1}(x^{(i)})}_{\checkmark} + \underbrace{\tfrac{1}{2}\alpha_j h_j(x^{(i)})}_{?}\right)}$$

$$= \sum \underbrace{\exp\left\{-y^{(i)} H_{j-1}(x^{(i)})\right\}}_{\checkmark} \underbrace{\exp\left\{-\tfrac{1}{2}\alpha_j y^{(i)} h_j(x^{(i)})\right\}}_{?}$$

$$= \sum \tilde{w}_j^{(i)} \exp\left\{-\tfrac{1}{2}\alpha_j y^{(i)} h_j(x^{(i)})\right\}$$

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$$\text{Exp-loss} = \sum_{i=1}^{N} \tilde{w}_j^{(i)} \exp\left\{ \sim\sim\sim\sim \right\}$$

$$= \sum_{y^{(i)} = h_j(x^{(i)})} \tilde{w}_j^{(i)} \exp\left\{ -\tfrac{1}{2}\alpha_j \cancelto{1}{y^{(i)} h_j(x^{(i)})} \right\}$$

$$+ \sum_{y^{(i)} \neq h_j(x^{(i)})} \tilde{w}_j^{(i)} \exp\left\{ -\tfrac{1}{2}\alpha_j \cancelto{-1}{y^{(i)} h_j(x^{(i)})} \right\}$$

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$$= e^{-\alpha_j/2} \sum_{\checkmark} \tilde{w}_j^{(i)} + e^{+\alpha_j/2} \sum_{\alpha} \tilde{w}_j^{(i)}$$

$$\pm e^{-\alpha_j/2} \sum_{\alpha}$$

finding optimal h_j

$$= e^{-\alpha_j/2} \sum_{\text{All}} \tilde{w}_j^{(i)} + \left(e^{+\alpha_j/2} - e^{-\alpha_j/2}\right) \sum_{\alpha} \tilde{w}_j^{(i)}$$

loss of AdaBoost

$$\sum_{\alpha} \tilde{w}_j^{(i)} = \sum_{\text{All}} \tilde{w}_j^{(i)} \mathbb{1}\left(y^{(i)} \neq h_j(x^{(i)})\right)$$

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$$\frac{\partial \text{Exp-loss}}{\partial \alpha_j} = 0 \rightarrow -\frac{1}{2} e^{-\alpha_j/2} \sum_{All} \tilde{w}_j^{(i)}$$

$$+ \frac{1}{2}\left(e^{\alpha_j/2} + e^{-\alpha_j/2}\right) \sum_{\alpha} \tilde{w}_j^{(i)}$$

$$= 0$$

$$\rightarrow e^{-\alpha_j/2} \sum_{All} = \left(e^{\alpha_j/2} + e^{-\alpha_j/2}\right) \sum_{\alpha}$$

$$\rightarrow \frac{e^{-\alpha_j/2}}{e^{\alpha_j/2} + e^{-\alpha_j/2}} = \frac{\sum_{\alpha}}{\sum_{All}} = \varepsilon_j$$

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$$e^{-\alpha_j/2} = x \longrightarrow \frac{x}{1/x + x} = \varepsilon$$

finding optimal α

$$\frac{x^2}{1+x^2} = \varepsilon$$

$$x^2 = \frac{\varepsilon}{1-\varepsilon} \longrightarrow x = \sqrt{\frac{\varepsilon}{1-\varepsilon}}$$

$$e^{-\alpha_j/2} = \sqrt{\frac{\varepsilon}{1-\varepsilon}}$$

$$\alpha_j = \ln\left(\frac{1-\varepsilon_j}{\varepsilon_j}\right) \checkmark$$

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$$\tilde{w}_j^{(i)} = \exp\left(-y^{(i)} H_{j-1}(x^{(i)})\right) \longrightarrow \text{in proof}$$

$$w_j^{(i)} = w_{j-1}^{(i)} \exp\left(\alpha_{j-1} \mathbb{1}\left(y^{(i)} \neq h_{j-1}(x^{(i)})\right)\right)$$

$\hookrightarrow$ in Alg.

$$= \underbrace{\exp\left(-y^{(i)} H_{j-2}(x^{(i)})\right)}_{\tilde{w}_{j-1}^{(i)}} \exp\left(-\tfrac{1}{2}\alpha_{j-1} y^{(i)} h_{j-1}(x^{(i)})\right)$$

$$= \tilde{w}_{j-1}^{(i)} \exp\left(-\tfrac{1}{2}\alpha_{j-1} y^{(i)} h_{j-1}(x^{(i)})\right)$$

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$$y\hat{y} = 1 - 2\mathbb{I}(y \neq \hat{y}) \longrightarrow \text{for } y, \hat{y} = \{+1, -1\}$$

$$= \tilde{w}_{j-1}^{(i)} \exp\left(-\tfrac{1}{2}\alpha_j\right) \exp\left(\alpha_j \mathbb{I}\left(y^{(i)} \neq h_{j-1}(x^{(i)})\right)\right)$$

Constant for all i

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